

1 General Order CC(n)

Based on the work of Kállay and Surján, the following in table 1 can be identified as possible diagram combinations. Note that I zero index the interactions to align with the conventions of the C programming language, as opposed to the Fortran programming language used by the authors of the original paper. Also, n will be the given residual level. The zeroth residual is the energy. All variables are limited by the highest excitation operator being considered.

Amplitude triplets				Interaction	Restrictions
0: f_{ab}					
X11	000	000	000	0	$X = n$
1: f_{ij}					
X10	000	000	000	1	$X = n$
2: f_{ia}					
X21	000	000	000	2	$X = n + 1$
X11	Y10	000	000	2	$X + Y = n + 1, X \geq Y$
X10	Y11	000	000	2	$X + Y = n + 1, X > Y$
3: $\langle ab cd \rangle$					
X22	000	000	000	3	$X = n$
X11	Y11	000	000	3	$X + Y = n, X \geq Y$
4: $\langle ij kl \rangle$					
X20	000	000	000	4	$X = n$
X10	Y10	000	000	4	$X + Y = n, X \geq Y$
5: $\langle ia jb \rangle$					
X21	000	000	000	5	$X = n$
X11	Y10	000	000	5	$X + Y = n, X \geq Y$
X10	Y11	000	000	5	$X + Y = n, X > Y$
6: $\langle ia bc \rangle$					
X32	000	000	000	6	$X = n + 1$
X22	Y10	000	000	6	$X + Y = n + 1, X \geq Y$
X10	Y22	000	000	6	$X + Y = n + 1, X > Y$
X21	Y11	000	000	6	$X + Y = n + 1, X \geq Y$
X11	Y21	000	000	6	$X + Y = n + 1, X > Y$
X11	Y11	Z10	000	6	$X + Y + Z = n + 1, X \geq Y \geq Z$
X11	Y10	Z11	000	6	$X + Y + Z = n + 1, X \geq Y > Z$
X10	Y11	Z11	000	6	$X + Y + Z = n + 1, X > Y \geq Z$
7: $\langle ij ak \rangle$					
X31	000	000	000	7	$X = n + 1$
X21	Y10	000	000	7	$X + Y = n + 1, X \geq Y$
X10	Y21	000	000	7	$X + Y = n + 1, X > Y$
X20	Y11	000	000	7	$X + Y = n + 1, X \geq Y$
X11	Y20	000	000	7	$X + Y = n + 1, X > Y$
X11	Y10	Z10	000	7	$X + Y + Z = n + 1, X \geq Y \geq Z$
X10	Y11	Z10	000	7	$X + Y + Z = n + 1, X > Y \geq Z$
X10	Y10	Z11	000	7	$X + Y + Z = n + 1, X \geq Y > Z$
8: $\langle ab ic \rangle$					
X11	000	000	000	8	$X = n - 1$
9: $\langle ai jk \rangle$					
X10	000	000	000	9	$X = n - 1$
10: $\langle ij ab \rangle$					
X42	000	000	000	10	$X = n + 2$
X32	Y10	000	000	10	$X + Y = n + 2, X \geq Y$
X10	Y32	000	000	10	$X + Y = n + 2, X > Y$

X31	Y11	000	000	10	$X + Y = n + 2, X \geq Y$
X11	Y31	000	000	10	$X + Y = n + 2, X > Y$
X22	Y20	000	000	10	$X + Y = n + 2, X \geq Y$
X20	Y22	000	000	10	$X + Y = n + 2, X > Y$
X21	Y21	000	000	10	$X + Y = n + 2, X \geq Y$
X22	Y10	Z10	000	10	$X + Y + Z = n + 2, X \geq Y \geq Z$
X10	Y22	Z10	000	10	$X + Y + Z = n + 2, X > Y \geq Z$
X10	Y10	Z22	000	10	$X + Y + Z = n + 2, X \geq Y > Z$
X21	Y11	Z10	000	10	$X + Y + Z = n + 2, X \geq Y \geq Z$
X21	Y10	Z11	000	10	$X + Y + Z = n + 2, X \geq Y > Z$
X10	Y11	Z21	000	10	$X + Y + Z = n + 2, X > Y > Z$
X10	Y21	Z11	000	10	$X + Y + Z = n + 2, X > Y \geq Z$
X11	Y21	Z10	000	10	$X + Y + Z = n + 2, X > Y \geq Z$
X11	Y10	Z21	000	10	$X + Y + Z = n + 2, X \geq Y > Z$
X20	Y11	Z11	000	10	$X + Y + Z = n + 2, X \geq Y \geq Z$
X11	Y20	Z11	000	10	$X + Y + Z = n + 2, X > Y \geq Z$
X11	Y11	Z20	000	10	$X + Y + Z = n + 2, X \geq Y > Z$
X11	Y11	Z10	W10	10	$X + Y + Z + W = n + 2, X \geq Y \geq Z \geq W$
X11	Y10	Z11	W10	10	$X + Y + Z + W = n + 2, X \geq Y > Z \geq W$
X11	Y10	Z10	W11	10	$X + Y + Z + W = n + 2, X \geq Y \geq Z > W$
X10	Y11	Z11	W10	10	$X + Y + Z + W = n + 2, X > Y \geq Z \geq W$
X10	Y11	Z10	W11	10	$X + Y + Z + W = n + 2, X > Y \geq Z > W$
X10	Y10	Z11	W11	10	$X + Y + Z + W = n + 2, X \geq Y > Z \geq W$
11: f_{ai}					
000	000	000	000	11	$n = 1$
12: $\langle ab ij \rangle$					
000	000	000	000	12	$n = 2$

Table 1: All possible diagram specifications for standard coupled-cluster. Here, n is the order of the residual, with $n = 0$ giving the variational energy, $n = 1$ giving the singles residual, and so on.

2 General Order CCn

For general order CCn, we use exact singles and for every amplitude, we expand up to the first time that T_n appears in its expression. In particular, the T_n amplitude is expanded to the $n - 1$ order, which is the first non-vanishing perturbative order. From here, we treat singles as zeroth order, as opposed to second order. This is equivalent to performing the T_1 transform, which has the side effect of making $\bar{F} = e^{-\hat{T}_1} \hat{F} e^{\hat{T}_1}$ first order, while $\hat{F}$ is zeroth order. This means that we get the resolvent for free, but we have to wait a level before the two-electron component of the $\bar{F}$ operator appears.

3 General Order LinCCn

For general order LinCCn, the restrictions are the same as for general order CC(n), but we only include diagrams with at most one amplitude. That is, for each operator in table 1, we only include the first entry, if possible.

4 General Order CC(n)(n + 1) and CC(n)[n + 1]

First, we need to define the amplitudes. The perturbative amplitudes are the same for both parenthesis and bracket methods. Note that $\hat{T}^{[1]} = \hat{T}_2^{[1]}$ while $\hat{T}^{[2]} = \hat{T}_1^{[2]} + \hat{T}_2^{[2]} + \hat{T}_3^{[2]}$ and for all others, $\hat{T}^{[n]} = \sum_{i=1}^{n+1} \hat{T}_i^{[n]}$.

$$t_I^{A[n]} = \left\langle \Phi_I^A \left| \left[\hat{V}, \hat{T}^{[n-1]} \right] + \sum_{\substack{1 \leq i \leq j \leq n-2 \\ i+j=n-1}} \left[\left[\hat{V}, \hat{T}^{[i]} \right], \hat{T}^{[j]} \right] + \sum_{\substack{1 \leq i \leq j \leq k \leq n-3 \\ i+j+k=n-1}} \left[\left[\left[\hat{V}, \hat{T}^{[i]} \right], \hat{T}^{[j]} \right], \hat{T}^{[k]} \right] \right. \right. \\ \left. \left. + \sum_{\substack{1 \leq i \leq j \leq k \leq l \leq n-4 \\ i+j+k+l=n-1}} \left[\left[\left[\left[\hat{V}, \hat{T}^{[i]} \right], \hat{T}^{[j]} \right], \hat{T}^{[k]} \right], \hat{T}^{[l]} \right] \right| \Phi_0 \right\rangle / \varepsilon_A^I \quad (1)$$

The result of this is that we need to generate the diagrams where the total perturbation is order n . We also need to consider that the $\hat{V}$ operator increases the perturbation by 1 and that $\hat{T}_1$ amplitudes anomalously increase the order by at least 2. To get the amplitudes used for the bracket and parenthesis methods, we need to replace the perturbative amplitudes with the converged amplitudes and remove repeated combinations. That is, $\hat{T}^{[1]} = \hat{T}_2$, $\hat{T}^{[2]} = \hat{T}_1 + \hat{T}_2 + \hat{T}_3$, and $\hat{T}^{[n]} = \sum_{i=1}^{n+1} \hat{T}_i$. Alternatively, we can consider all combinations of amplitudes and the $\hat{V}$ operator where the amplitudes are treated as if they are their lowest possible order, the $\hat{V}$ operator is first order, and the total order must be less than or equal to n , as opposed to simply being equal to n . This latter version is much easier to implement, as we don't need to consider the cases where an amplitude is included at a higher perturbation level than expected.

Once we have the amplitude expression, we can figure out the energy. We'll start with the bracket energy.

$$\Delta E_{CC(n)[n+1]} = \langle \Phi_0 | \hat{T}_2^\dagger \hat{V} \hat{R}_4^{[n+1]} | \Phi_0 \rangle + \begin{cases} 0 & n \equiv 0 \pmod{2} \wedge n > 2 \\ \langle \Phi_0 | \hat{T}_3^\dagger \hat{V} \hat{R}_5^{[n+1]} | \Phi_0 \rangle & \text{else} \end{cases} \quad (2)$$

Here, we define the $\hat{R}_m^{[n+1]}$ as follows.

$$\hat{R}_m^{[n+1]} = \begin{cases} \sum_{AI} \frac{\langle \Phi_I^A | \hat{V} (\hat{R}_{m+2}^{[n+1]} + \hat{R}_{m+1}^{[n+1]}) | \Phi_0 \rangle}{\varepsilon_I^A} \tilde{a}_I^A & m \equiv 0 \pmod{2} \wedge n \equiv 0 \pmod{2} \wedge m < n \\ \sum_{AI} \frac{\langle \Phi_I^A | \hat{V} \hat{R}_{m+2}^{[n+1]} | \Phi_0 \rangle}{\varepsilon_I^A} \tilde{a}_I^A & m \leq n \\ T_{n+1}^{[n]} & m > n \end{cases} \quad (3)$$

where A and I are lists of m virtual or occupied indices, $\varepsilon_I^A = \sum_{i \in I} \varepsilon_i - \sum_{a \in A} \varepsilon_a$ is the resolvent, built out of the Hartree-Fock orbital energies, and $\tilde{a}_I^A = \prod_{a \in A} a_a^\dagger \prod_{i \in I} a_i$ is a normal-ordered excitation operator. Next, the parenthesis energy.

$$\Delta E_{CC(n)(n+1)} = \langle \Phi_0 | \hat{T}^{[1]\dagger} \hat{V} \hat{R}_4^{(n+1)} | \Phi_0 \rangle + \langle \Phi_0 | \hat{T}^{[2]\dagger} \hat{V} \hat{R}_5^{(n+1)} | \Phi_0 \rangle \quad (4)$$

Here, we define the $\hat{R}_m^{(n+1)}$ in almost exactly the same way as $\hat{R}_m^{[n+1]}$, only we don't care about the parity of n . Explicitly,

$$\hat{R}_m^{(n+1)} = \begin{cases} \sum_{AI} \frac{\langle \Phi_I^A | \hat{V} (\hat{R}_{m+2}^{(n+1)} + \hat{R}_{m+1}^{(n+1)}) | \Phi_0 \rangle}{\varepsilon_I^A} \tilde{a}_I^A & m \equiv 0 \pmod{2} \wedge m < n \\ \sum_{AI} \frac{\langle \Phi_I^A | \hat{V} \hat{R}_{m+2}^{(n+1)} | \Phi_0 \rangle}{\varepsilon_I^A} \tilde{a}_I^A & m \leq n \\ T_{n+1}^{[n]} & m > n \end{cases} \quad (5)$$

This means we have a few special cases. For CCSD(T), the energy expression gives the following.

$$\Delta E_{CCSD(T)} = \langle \Phi_0 | \hat{T}_2^\dagger \hat{V} \hat{T}_3^{[2]} | \Phi_0 \rangle + \langle \Phi_0 | \hat{T}_1^\dagger \hat{V} \hat{T}_3^{[2]} | \Phi_0 \rangle \quad (6)$$

Otherwise, when n is even, $\Delta E_{CC(n)[n+1]} = \Delta E_{CC(n)(n+1)}$, and when n is odd, the $\Delta E_{CC(n)[n+1]}$ expression is included in the $\Delta E_{CC(n)(n+1)}$ expression. Also, when n is odd, the $\Delta E_{CC(n)[n+1]}$ expression only contains a single term. Note that while this recurrence relation is written bottom-up, a good implementation will use the fact that the top-down approach can be factored to better reuse intermediates.

4.1 Lambda Corrections

The $\hat{T}^\dagger$ used in the parenthesis are only an approximation for the $\hat{\Lambda}$ operators. If we use the converged $\hat{\Lambda}$ operators instead, we get the CC(n)(n + 1)_Λ methods. Note that the bracket expansion does not try to approximate the $\hat{\Lambda}$ operators, so we shouldn't try to replace them there. If we did, it would give the CC(n)(n + 1)_Λ equations right back, meaning a hypothetical CC(n)[n + 1]_Λ would be exactly the same as CC(n)(n + 1)_Λ for all n , not just even n . Also note that there are no $\hat{\Lambda}$ equations for the perturbative amplitudes, meaning we can't go much further from here. This is because no amplitudes depend on the $\hat{T}^{[n]}$ operators, making it impossible to take a derivative by the $\hat{T}^{[n]}$ operators to generate the $\hat{\Lambda}^{[n]}$ residual. For property calculations, we can substitute $\hat{\Lambda}^{[n]} = \hat{T}^{[n]\dagger}$, but we won't capture any orbital relaxation effects due to the higher-order excitations. This also means that we can't perform EOM-CC(n)(n + 1) or EOM-CC(n)[n + 1]